AS MATHEMATICS - MECHANICS 1 - DAY 13

M1 consolidation

A full mixed checkpoint

YOUR AIM TODAY

Complete a closed-notes checkpoint and expose the next priority topic.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

The ideas you need

A mechanics solution earns clarity from its diagram, equation, direction and units.

- Draw a force or motion sketch.
- Declare a positive direction.
- Write the model equation symbolically before substituting.
- Use g=9.8 m/s^2 unless told otherwise.
- Target: five fully correct solutions.

MEMORY RULE

Diagram, direction, equation, solve, interpret.

Exam habit

WORKED STEPS

If acceleration is negative, say whether the particle is slowing or accelerating in the negative direction.

Correction habit

WORKED STEPS

Classify errors as diagram, sign, formula, algebra or units.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. A car goes from 5 to 17 m/s in 4 s uniformly. Find a and s.	
2. A 6 kg block has 30 N forward and 12 N resistance. Find a.	
3. Find limiting friction for 8 kg on horizontal ground, mu=0.25.	
4. A 2 kg body rises 3 m. Find GPE gained.	
5. A 10 kW engine at 5 m/s balances resistance. Find resistance.	
6. 2 kg at 8 m/s sticks to 6 kg at rest. Find final speed.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	a=3 m/s^2; s=(5+17)4/2=44 m.
2	Resultant 18 N; a=3 m/s^2.
3	R=8g; F=0.25(8g)=19.6 N.
4	mgh=2g(3)=58.8 J.
5	F=P/v=10000/5=2000 N.
6	16=8v, so v=2 m/s.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.